

Operationalising Observational Instability in Misinformation Systems

A Human-Guided Early-Warning Framework

PhD Research Proposal

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Research Area: Computational Social Science, Misinformation, Network Science, and Machine Learning

1 Project Summary

Contemporary misinformation detection draws on content, propagation, and behavioural or user-centred signals to determine whether information is misleading or false. However, the models and evidence-backed reference records supporting these judgements are developed under particular conditions. As available evidence, claim formulations, and communication environments change, the basis for an existing judgement may require reconsideration. The challenge is therefore not only whether a detector makes an isolated error, but whether the wider observation system remains adequate under changing conditions.

This project will develop and empirically evaluate a human-guided early-warning framework that extends existing misinformation detection systems rather than replacing them. The project uses the term *observational instability* to describe situations in which the basis for an existing misinformation observation may require review. Rather than issuing another true/false verdict, the proposed supervisory layer will identify when such review may be necessary. A human reviewer will then determine whether the computational model, the evidence-backed reference record, both, or neither should be updated, with any revision informing the next observation cycle.

Through three connected studies, the project will identify and examine the indicators and mechanisms of observational instability, construct a dynamic longitudinal claim–evidence dataset, and evaluate whether warning-triggered human-guided updating can maintain more reliable observation than static or periodically updated approaches.

2 Background and Research Gap

Misinformation detection systems are commonly developed and evaluated against predefined labels and benchmark datasets. However, performance within a fixed evaluation setting may provide limited evidence of reliability under changing real-world conditions. Benchmark contamination can inflate the apparent performance of LLM-based detectors, while dataset shift—where the data encountered after deployment differ systematically from those used for model development—can reduce both predictive performance and the reliability of model uncertainty (Xu & Yan 2025, Ovidia et al. 2019).¹ A model may therefore perform well against an established benchmark without

¹For example, a detector trained mainly on long-form political claims may later encounter short-form health claims on a different platform. Although the classification task remains the same, differences in topic, vocabulary, sources, and communication style may make patterns learned from the original data less reliable. This is one form of dataset shift. Recent work suggests that such shifts can be partly mitigated by few-shot domain adaptation methods, which use a limited number of labelled target-domain examples to guide knowledge transfer from the source domain (Yue et al. 2023). However, their effectiveness depends on access to suitable target-domain examples, and they primarily adapt the computational model rather than assess whether the wider observation system remains adequate as evidence and communication conditions continue to change.

demonstrating that its learned representations remain appropriate beyond that environment.

Two forms of change are particularly important. First, the claim may remain unchanged while the evidence available for judging it develops over time. For example, LiveFact, a time-aware benchmark that evaluates claims against the evidence available at successive stages of unfolding events, shows that a judgement justified at one point in time may differ from the claim’s final verification label (Xu et al. 2026).² Second, the evidence and underlying claim may remain largely unchanged while the claim is expressed differently. Meaning-preserving reformulations and style changes can nevertheless alter classifier decisions (Wu et al. 2024, Przybyła et al. 2025).³ These cases imply different update needs: the first may require revision of the evidence-backed reference record, whereas the second may indicate that the computational model requires updating.

Related research addresses parts of this problem. Concept-drift methods adapt predictive models when relationships in incoming data change, while adaptive verification systems retrieve additional evidence before returning a judgement (Gama et al. 2014, Xie et al. 2025). Research on automated moderation also demonstrates the continuing need for human configuration and contextual interpretation (Jhaver et al. 2019). However, these approaches mainly adapt a model, gather further evidence, or check an individual output. They do not jointly determine whether a problem arises from the model, the reference record, or the relationship between them.

The literature therefore leaves three connected questions unresolved. First, it remains unclear which measurable changes indicate that an existing misinformation observation may require review. Second, it is not known whether such changes can provide reliable warning before a model failure or reference-record revision becomes apparent. Third, existing research offers limited evidence on whether human-guided review can identify the appropriate target of updating—the computational model, the evidence-backed reference record, both, or neither—and whether such targeted updating is more effective than static or periodically updated approaches. These unresolved issues motivate the project’s investigation of *observational instability* as a possible basis for early warning and targeted review.

3 Research Aim and Questions

3.1 Research Aim

To address this gap, this project aims to develop and empirically evaluate a human-guided early-warning framework for identifying *observational instability* and supporting targeted updating of misinformation observation systems.

3.2 Main Research Question

RQ: How can observational instability in misinformation observation systems be identified early and used to guide human review and targeted updating under changing communication conditions?

²Suppose a claim posted during a breaking event states that “20 people have died”. At the time of posting, official sources may not yet have confirmed any casualty figure, so the claim cannot be reliably verified. If authorities later confirm 20 deaths, the same claim can then be supported by the available evidence. What has changed is not the wording of the claim, but the evidential basis on which it is judged.

³For example, Przybyła et al. demonstrate that a sentence such as “we failed to recall that they know nothing at all about catholicism” can be reformulated as “we forgot that they don’t have a clue about catholicism”. The meaning is largely preserved, but such reformulations can be used to change the decision of a credibility classifier. In this case, the relevant evidence has not changed; the problem lies in the model’s sensitivity to how the information is expressed.

3.3 Sub-questions

SubRQ1: What measurable changes in model behaviour, available evidence, and communication conditions indicate that an existing misinformation observation may require review?

SubRQ2: Can these signals provide reliable early warning before the current observation becomes demonstrably unreliable or requires retrospective correction?

SubRQ3: Once a warning is triggered, can human review correctly identify the appropriate update target—the computational model, the evidence-backed reference record, both, or neither—and does such targeted updating maintain more reliable observation than static or periodically updated approaches?

4 Research Design and Expected Outputs

The research design is grounded in a distinction between judging whether a claim constitutes misinformation and assessing whether the system producing that judgement remains adequate. A misinformation observation is understood as the joint outcome of a computational model, an evidence-backed reference record, the evidence available at a particular time, and the communication conditions under which the judgement is produced. The project terms potential misalignment among these elements *observational instability*.

The proposed framework adds a supervisory layer to existing misinformation detection systems. When possible instability is identified, the system alerts a human reviewer, who determines whether the model, the reference record, both, or neither require updating. Any resulting revision informs the next observation cycle. The framework will be operationalised through three connected studies over a three-year research programme.

4.1 Study 1: Identifying Indicators and Mechanisms of Observational Instability

Study 1 will use a theory-informed comparative design to identify what changes may indicate that an existing misinformation observation requires review and how such instability may emerge. Candidate indicators will be derived from prior experimental research and examined using existing datasets such as LiveFact (Xu et al. 2026), AVeriTeC (Schlichtkrull et al. 2023), TSVer (Strong & Vlachos 2025), and PHEMEPlus (Dougriez-Lewis et al. 2022).

Temporal changes in available evidence will be used to examine cases in which the reference judgement may require revision, while meaning-preserving reformulations will test cases in which the reference judgement remains stable but model behaviour changes (Wu et al. 2024, Przybyła et al. 2025). Study 1 will identify which changes in evidence, model behaviour, and communication conditions are useful signals that an existing observation requires review. These findings will determine what information is recorded longitudinally in Study 2.

4.2 Study 2: Constructing a Dynamic Longitudinal Claim–Evidence Dataset

Guided by the data requirements and candidate indicators identified in Study 1, Study 2 will construct a new domain-specific longitudinal claim–evidence dataset. Existing research resources used in Study 2 will primarily provide design references, seed records, and comparative material rather than being directly merged. The main longitudinal records will be reconstructed from claim and evidence sources, such as professional fact-checking repositories, official records and announcements, and news archives, together with communication-platform data where relevant and accessible. All sources will be subject to appropriate access, licence, reuse, and ethics conditions. Figure 1 summarises the planned construction process.

Related claim formulations will be organised into claim clusters and reconstructed across multiple

observation points. At each point, circulating claim formulations will be aligned with the evidence available at that time, while timestamps and source provenance are preserved. Communication-platform records will provide additional information on how claims are reformulated, circulated, and interacted with over time. Where temporal or evidential gaps remain, additional time-stamped evidence will be collected from approved sources. This process will produce longitudinal records linking claim formulations, available evidence, communication activity, observation time, and provenance.

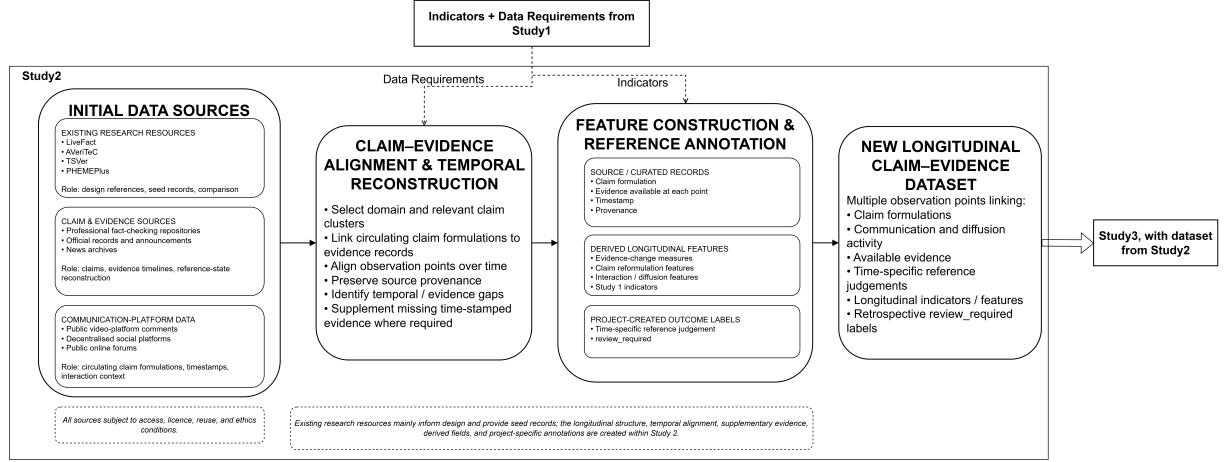


Figure 1: Planned workflow for constructing the longitudinal claim–evidence dataset in Study 2. Data requirements and candidate indicators identified in Study 1 guide temporal reconstruction and feature construction, while subsequent validated evidence supports the creation of retrospective reference annotations.

The resulting records will then be used to construct longitudinal features that operationalise the candidate indicators identified in Study 1, including changes in available evidence, claim formulation, and communication or diffusion activity. A time-specific, evidence-backed reference judgement will be established for each observation point. Subsequent validated evidence will then be used retrospectively to determine whether the earlier reference judgement required material revision. If the later evidence materially conflicts with, qualifies, or supersedes the earlier reference state, the corresponding observation will be labelled `review_required = yes`; otherwise it will be labelled `review_required = no`. The criteria for what constitutes a material revision will be developed and validated through Studies 1 and 2.

The resulting dataset will therefore contain the information available at each observation point together with a retrospectively constructed `review_required` outcome. Evidence appearing after time t will be used only to establish this target and will not be available to the early-warning system at time t . Study 3 will use the dataset to test whether future review requirements can be anticipated from contemporaneous evidence, communication and diffusion signals, and the indicators identified in Study 1.

4.3 Study 3: Developing and Evaluating a Human-Guided Early-Warning System

Study 3 will use the longitudinal dataset constructed in Study 2 to train and evaluate an early-warning model for predicting the retrospective `review_required` label. The model will use the indicators identified in Study 1 and, at each observation point, will be restricted to information available at time t , including the current claim formulation, available evidence, and communication and diffusion signals. This design tests whether the need for review can be anticipated before a later reference revision becomes apparent.

During deployment, whenever the early-warning model predicts `review_required = yes`, a human agent will fact-check the corresponding reference record. If the review identifies a conflict between the record and the current ground truth, the record will be updated. Otherwise, the record will remain unchanged and a cumulative model-unreliability score will be increased. This score will decay over time, reducing the influence of older review outcomes. Reaching a specified threshold will be used as the criterion for triggering an update to the early-warning model, either through retraining or modification of the model. The scoring rule, decay mechanism, and threshold will be developed and evaluated during the project. Updated records and models will inform subsequent observation cycles.

The warning-triggered approach will be compared with static and fixed-interval updating approaches. Evaluation will address SubRQ2 by examining warning accuracy and timeliness, and SubRQ3 by assessing the appropriateness of reference-record and early-warning model updates, the reliability of subsequent observations, and the amount of human review required. Building on approaches to evidence retrieval and human review (Xie et al. 2025, Jhaver et al. 2019), the study will connect early warning, human diagnosis, targeted updating, and subsequent observation within a single feedback process. Future evidence will not be available to the system at earlier observation points. Candidate data sources have been preliminarily assessed for access, licence, reuse, and redistribution conditions, while human-review procedures will follow the required ethics process.

5 Significance and Contributions

The significance of this project lies not in producing another misinformation detector, but in extending misinformation research from one-off classification towards the continuing review and updating of observations under changing conditions. The project also has relevance to natural language processing, adaptive machine-learning systems, and human-in-the-loop design (Mosqueira-Rey et al. 2022).

5.1 Reframing Misinformation Detection as Adaptive Observation

For misinformation research and natural language processing, the project will extend the task from producing one-off claim-level verdicts towards monitoring whether the basis of an existing judgement remains adequate over time. Conventional early detection generally asks whether misinformation can be identified before it spreads widely. By contrast, the proposed early-warning task asks whether an existing observation may require review before it becomes demonstrably unreliable.

This reframing introduces two related tasks: determining whether a current observation requires review through `review_required`, and identifying the appropriate target of updating through `update_target`. It also supports a temporal and evidence-aware form of NLP evaluation in which claims, evidence, reference judgements, and model outputs are treated as versioned rather than permanent. This will make it possible to distinguish what could reasonably be concluded at a particular time from what became known only retrospectively.

5.2 Event-Triggered Adaptation and System-Level Resilience

For a system perspective, the project will apply an event-triggered adaptation logic to misinformation observation. Rather than assuming that systems should never update, update at fixed intervals, or retrain whenever environmental change is detected, the proposed framework will use signs of observational instability to determine when review should be activated. The subsequent response may involve updating the model, the evidence-backed reference record, both, or neither.

This approach extends adaptation beyond model parameters alone and treats the model and its reference records as jointly situated within a changing environment. This challenge also reflects a form of *epistemic instability*, referring to situations in which the continued reliability of an observation process becomes uncertain as the underlying model, reference record, or evidential conditions change.⁴ Identifying when review is needed is distinct from determining what has become unreliable. The project therefore examines whether early warning can support targeted intervention and maintain reliable observation without unnecessary human review.

5.3 Human Agency as an Interpretive Anchor in Human–AI Systems

The project will examine human agency as a situated and revisable interpretive anchor within the observation system. Following an early warning, human reviewers will interpret the flagged observation and diagnose the source of the instability identified in Section 5.2. Importantly, this diagnosis does not assume that instability originates from the computational model alone; it requires consideration of whether the issue arises from the model, the evidence-backed reference record, or changing communication and evidential conditions. This extends human-in-the-loop design beyond routine annotation or validation towards contextual diagnosis under changing conditions (Mosqueira-Rey et al. 2022).

This role does not assume that human judgement provides direct or infallible access to reality. Human interpretation remains situated and open to revision. Selective human review will allow the project to examine when intervention improves subsequent observation and when it adds unnecessary disagreement or workload. By combining automated analysis with human agency, the project provides a contemporary empirical case of how automated systems and human agency can be coordinated to manage uncertainty in real-world classification tasks. This coordination extends beyond the immediate review, as any resulting intervention informs the next observation cycle.

6 Research Preparation and Fit with the Research Environment

This project builds on my Master’s research at The Australian National University⁵ and my training in Economic Statistics, Computing, Cybernetics, and network science. I plan to combine NLP-based measures of potential conflict between circulating claims and evidence-backed reference records with network models of information diffusion to develop an early-warning system. When warning alarms, human agent will act as interpretive anchors, reviewing alerts and deciding whether models or reference records need updating as evidence and communication conditions change.

Using misinformation diffusion and the circulation of review alerts as a case study, I hope to explore how people and AI can work together to support more resilient social systems. This connects with the Resilience Engineering program’s focus on maintaining and recovering system functions under disturbance and adapting to change. Here, resilience means recognising the limits of existing judgements and supporting their review and revision.

⁴Epistemic uncertainty in machine learning concerns uncertainty arising from incomplete knowledge about the underlying model or data-generating process (Hüllermeier & Waegeman 2021). Unlike uncertainty about a current state of knowledge, the notion of instability emphasises how the reliability of knowledge production or evaluation may change as the underlying system or conditions evolve (Kardong-Edgren 2026). In this project, epistemic instability is adapted to misinformation observation systems, describing situations where the basis for an observation may no longer remain reliable without immediately revealing whether revision is required for the computational model or the evidence-backed reference record. For example, later validated evidence may require revision of an earlier reference judgement, whereas the reference judgement may remain stable while a meaning-preserving reformulation changes the model’s prediction.

⁵My Master’s thesis was titled *Reconsidering Misinformation within Adaptive Socio-Technical Communication Environments: A Conceptual Exploration*.

At the Laboratory scale, Toriumi Laboratory's work on social media, complex networks, and information diffusion closely matches my research interests. I believe that Professor Toriumi's expertise, together with the laboratory's research focus on the analysis and design of social media and web services, as well as complex network science, would help me develop this project further. In turn, I hope to contribute a complementary perspective from my training at the ANU School of Cybernetics, particularly on how feedback and human agency can support resilience in socio-technical systems.

Appendix A: Illustrative Data Sources and Reusable Design Features

This appendix provides a preliminary assessment of data sources that may support the construction of the proposed longitudinal claim–evidence dataset. Their inclusion does not represent a final data-selection decision. Existing research resources will primarily serve as design references, seed records, and comparative material, while additional domain-specific records may be constructed from claim and evidence sources and, where relevant and accessible, communication-platform data. Final source selection and reuse will be subject to the data requirements identified in Study 1 and to source-specific access, licence, redistribution, and ethics conditions.

Controlled robustness resources, such as TREPAT, will be considered separately in Study 1. Such resources support model-side stress testing through meaning-preserving claim reformulation, but do not themselves provide the longitudinal claim–evidence records required for Study 2 ([Przybyła et al. 2025](#)).

Resource	Available Data and Structure	Reusable Features and Key Limitations	Access, Licence, and Reuse Conditions
LiveFact Dataset Repository (Xu et al. 2026)	Claims are paired with evidence made available at different stages of unfolding news events. The benchmark distinguishes final classification from the judgement supported by the evidence available at a particular time.	Reusable feature: temporal evidence snapshots and the distinction between an eventual factual outcome and a judgement that was justified at an earlier observation point. Limitation: it is primarily a controlled, news-event benchmark and does not provide a complete record of real-world platform interaction, propagation, or human updating.	Publicly accessible through Hugging Face. The dataset is released under the Apache License 2.0 , permitting reuse, modification, and redistribution subject to the licence conditions. Rights associated with any third-party source material will be checked separately before redistribution.
AVeriTeC Dataset and code Paper (Schlichtkrull et al. 2023)	Real-world claims from professional fact-checking organisations are linked to traceable web evidence, structured question-answer records, verdicts, and written justifications.	Reusable feature: evidence provenance, structured evidence-backed reference records, and explanations of how evidence supports a verdict. Limitation: records are principally designed for claim-level verification rather than repeated observation of the same claim across multiple time points.	Publicly accessible through the project repository and released under CC BY-NC 4.0 . Non-commercial reuse, adaptation, and redistribution are permitted with attribution. Linked or archived external web sources will be treated as third-party material and checked separately before reuse or redistribution.
TSVer Dataset and code Paper (Strong & Vlachos 2025)	Real-world claims are paired with historical time-series evidence, relevant time ranges, verdicts, metadata, and written justifications.	Reusable feature: time-bounded evidence and reference judgements, which may inform version fields such as <code>valid_from</code> , <code>valid_to</code> , and temporal revision reasons. Limitation: its emphasis on numerical and time-series claims may not capture broader narrative, behavioural, or platform-level forms of misinformation.	Publicly accessible through GitHub and released under CC BY-SA 4.0 . Reuse, adaptation, and redistribution are permitted with attribution, while redistributed adaptations are subject to the ShareAlike requirement. The dataset preserves provenance for its underlying time-series evidence.
PHEMEPlus Dataset and code Paper (Dougrez-Lewis et al. 2022)	Labelled social-media rumours and their conversation threads are connected with relevant external web evidence.	Reusable feature: a structure linking claim clusters, conversation or interaction context, and external evidence. This may support the communication-side indicators identified in Study 1. Limitation: it is based on historical platform data, and some external evidence must be retrieved again from recorded URLs rather than being redistributed directly.	The dataset is publicly downloadable, but the project repository does not state an explicit licence for PHEMEPlus. Its underlying PHEME annotations are available under CC BY 4.0, while social-media content and external evidence remain subject to separate rights. The PHEMEPlus authors provide URLs rather than redistributing external evidence because of copyright restrictions. Reuse or redistribution will therefore be limited to components for which permissions are clear, with further clarification sought if necessary.
New Domain-Specific Records	Additional longitudinal records may be constructed from professional fact-checking repositories, official records and announcements, news archives, and, where relevant and accessible, public communication-platform data.	Reusable feature: these records can fill temporal, evidential, and contextual gaps in existing research resources and support the reconstruction of claim formulations and evidence available at multiple observation points. Limitation: construction will require claim matching, temporal alignment, evidence archiving, provenance tracking, and project-created annotation.	Access and reuse conditions will be assessed separately for each source. Raw third-party or platform content will not be redistributed where source terms or copyright do not permit it; where necessary, the project will retain source identifiers, URLs, metadata, or derived annotations instead. Collection and reuse will also be subject to applicable platform conditions and ethics requirements.

Table 1: Illustrative data sources, reusable design features, limitations, and access and reuse conditions relevant to the proposed longitudinal dataset.

Appendix B: Illustrative Reference-Side Early-Warning and Update Pathway

This appendix presents one illustrative conceptual pathway through which an evidence-backed reference record may be monitored and reviewed. It is included to demonstrate the system-update logic of the proposed framework rather than to prescribe a final computational architecture.

In this example, the current evidence-backed reference state A_t is challenged by newly available evidence or related informational events. Possible challenge, influence, propagation, and disturbance factors are aggregated to assess whether the existing reference state may require reconsideration. When the resulting assessment crosses a review threshold, the system generates an early warning for human review. The warning does not establish that the reference record is incorrect; it indicates only that its continued use may require renewed examination.

The human reviewer then considers the available evidence and contextual conditions and determines whether the reference record should be retained or revised. Any accepted revision produces an updated reference state A_{t+1} , which becomes the basis for the next observation cycle.

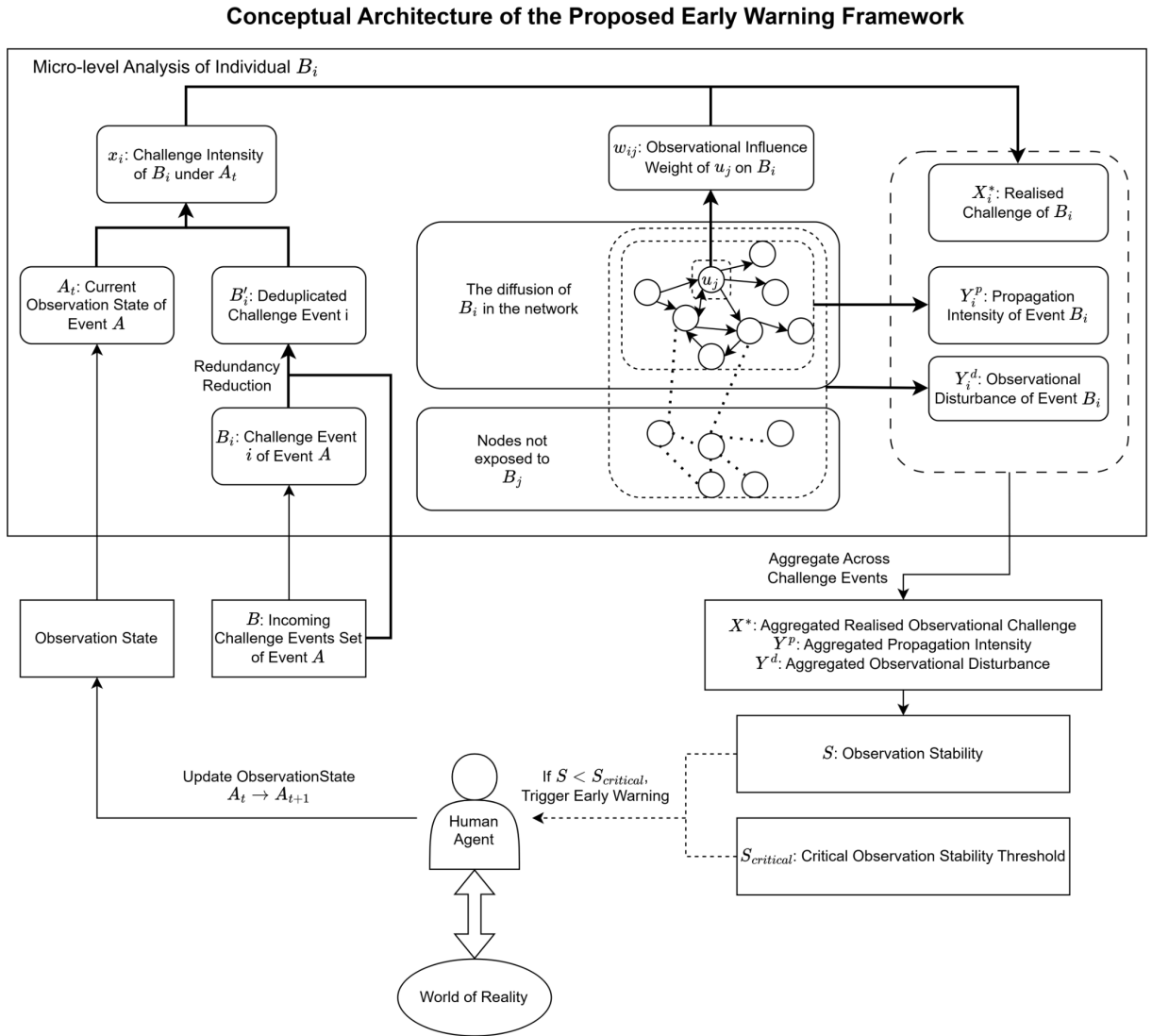


Figure 2: Illustrative conceptual example of a reference-side early-warning and update pathway.

Note. The factors, functional relationships, aggregation procedures, and thresholds shown

in Figure 2 are unvalidated conceptual placeholders. They illustrate possible components of a reference-side warning process and will be revised, replaced, or removed according to the operationalisation and empirical testing conducted in Study 1. The diagram represents only a possible REFERENCE_RECORD_ONLY pathway; model-side, joint-update, and no-update outcomes are not represented.

Source. Adapted from the conceptual early-warning framework developed in the author’s unpublished Master’s thesis, *Reconsidering Misinformation within Adaptive Socio-Technical Communication Environments: A Conceptual Exploration*.

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